

ActInf GuestStream 097.1 ~ Matthew Sims: Slime Mould and Philosophy

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Hello, welcome everyone. It's January 29th, 2025. We're in Active Inference Guest Stream 97.1 with Dr. Matthew Sims discussing Acellular Slime Mold, a jumping off point for exploring different notions of biological memory. So, thank you for joining. Looking forward to hearing more.

Thanks so much. Hello, welcome everyone. It's January 20, 2025. We're in the next one.

Returning in one second. Thank you.

We're back. Like a slime mold. Returning. We're back.

Shall I start? Or should I? Shall we just go for it?

Yeah, yeah. Begin from the beginning of the presentation. Thank you.

I can't hear you.

Oh, thank you. We're back. Go for it.

Okay. Sorry about that. So yeah, today I'm going to be looking at a couple of questions regarding *Phauserum Polycephalum*, which is Acellular Slime Mold. And I'm going to be looking at different notions of memory that have been thought about in the biological sciences, not necessarily in, for example, when you're thinking about large language models or when you're thinking about computational memory outside of biological systems, but biological memory. *Phauserum Polycephalum* is a kind of, is a springboard, if you will, for thinking about things like this. Now this organism, I'm going to get a little bit into its biology and a little bit into kind of its, its complex life cycles. And these are, these are different things that I kind of have researched and thought about in the kind of more recent book that I've published with Cambridge University Press called *Slymold and Philosophy*. And this particular talk that you'll see right now is going to be kind of an excerpt, if you will, from a couple of the chapters, namely chapter three of the book.

And looking at memory, I think that's a quite nice way to get, to get it, to get a kind of an overall understanding of the importance of this, this system, this model system. And it's a system that's becoming used more and more to investigate various questions in either cognitive science or in ecology or in, or in philosophy, biology or theoretical biology, more generally. Questions like biological individuality, for example, questions like, you know, the biological or the biological status of something like a spore, dormant spore, when we're thinking about living systems being systems that are metabolically active. So what does that mean if life is, you have a form of life that's not metabolically active and looking at the biology of something like a pizarin polycephaline or slime mold can really, I think it's very kind of instrumental into digging into these questions, more theoretical questions. And also you can explore and expand upon these questions by generating hypotheses that can be really tested in an organism that's quite easy to culture. So this talk is going to proceed as follows with that kind of background. I'm going to look at what pizarin is, then I'm going to turn to a really, really, really beautiful experiment. It's a very interesting and powerful experiment, well designed by Chris Reed

and colleagues at Macquarie University. They're still, they're still doing experiments with pizarin. And this particular experiment is about kind of externalized memory in pizarin. I'll look at that experiment, go through it, then I'll look at different forms of biological memory. And these different forms of biological memory, I'm going to cap it out at three or four different forms of biological memory and try to connect them all. And then I'm going to ask a question regarding whether or not one or more of these forms of biological memory requires an explanation using kind of cognitive theories or cognitive frameworks. And I'm going to answer that question using the notion of metabolic individuality. I said, again, it's biological individuality, which we'll get into during the talk also. And so applying this particular type of criterion from biological individuality or metabolic individuality as a criterion, I'm going to be able to kind of rule out which levels of memory, depending upon the levels of biological organization that this memory happens in, are, you know, the warranted targets of cognitive explanation. So what's Pizarin Polycephalum? So Pizarin Polycephalum, it's the Latin name, the genus is Pizarin, and Polycephalum is the kind of the species name, the species add-on. And that generally means, you know, many-headed. And they say there's many-headed in the sense that it's a many, the spore comes out of what looks like to be a many-headed fruiting body. Generally, it's an a-cellular slime mold. It's a protist, a unicellular protist. And it has a very interesting complex life cycle. And when I say complex life cycle, what I mean is that it has, it has multi-facets, it's multi-faceted in the sense that it has more than one vegetative stage, one than more stage where it actually has a form that reproduces and kind of, you know, forges, eats. And this stage goes from a spore at number one, the spore actually germinates, and out from the spore, you have something called a biflagellate cell or a mixed amoeba cell. Now, a mixed amoeba comes out of the spore, it germinates from the spore. When there's less water in the environment, the biflagellate actually is something that is the kind of the flagella are going to be something that's synthesized when there's no water in the area when it germinates. Now, during a mixed amoeba stage, it can reproduce asexually. So there are many, many different kind of clones, the clone cells, and each of these have a single nucleus. When it actually encounters another type, another mixed amoeba of the same type, not necessarily genetically the same, they can actually come together. And that's when you have, that's when you have a type of reproduction that occurs. It's mitosis without actually cell separation, so without cytokinesis. And that's when you start having this particular growth of a cell. And these are all microscopic small single cells until this stage where the nuclei reproduce and it starts growing and becomes visible to the eye. And this is, becomes a plasmodial stage. Now this plasmodial stage is the stage at which most of the experiments with physarium occur. And it's a very interesting stage because this, at this stage, the physarium, the cells, again, it's a single cell, multinucleate single cell, crawls over and locomotes over various surfaces. And you can design experiments watching this particular type of locomotion. Now, again, I talked about the vegetative stage. I've said pretty much this, during this vegetative stage, during this plasmodial stage, it's going to consume bacteria, fungi, and decaying organic matter. And again, here you have a picture from one of my lab experiments. I'm just locomoting and covering oats. During this plasmodial

stage, or the vegetative stage, one of its vegetative stages, there's again, many experiments that have shown interesting types of capacities that it displays. It's been shown to kind of solve the shortest path problem, engage in habituation, which is a type of learning, very simple type of learning, anticipate periodic environmental conditions, hot bursts of air, dry bursts of air, excuse me, and make flexible behavioral decisions that are based upon comparative evaluation rules rather than absolute evaluation. And this is just, you know, a very, very brief summary of the various experiments that have been, that *Fascerum* has been used with to engage in questions regarding things like learning, things like decision-making, and there are many more experiments, but you get a kind of representation of what this might be. Now, let's move on with that kind of brief understanding of what, you know, the life cycle of *Fascerum* is and focusing upon this particular type of vegetative stage, the plasmodial stage. We can start thinking about one experiment in particular, the experiment by Chris Reed and his lab over at McCreary in 2012. Now, you'll see a particular, you know, picture here of Hansel and Gretel, and I hope to make this clear why I'm showing this to you. So when *Fascerum* moves, when it locomotes over surfaces, it leaves an extracellular slime trace or slime trail. I'll refer to this as ES. Now ES is, it's a polymucosaccharide, and it's a, this, this, this mucus makes it easier for *Fascerum* to actually locomote, but it serves many, many functions. And it's been well known in the literature since the fifties that wherever *Fascerum*, you know, wherever it navigates, wherever it locomotes, it leaves this trail behind it. So one question that has risen, and a question that is kind of the crux of the experiment for Reed et al. is whether or not *Fascerum* uses this extracellular slime to navigate. So the reason why they can ask this question, that has been observed that *Fascerum* avoid their own extracellular slime trails and the extracellular slime trails of others. Now, if it's used as a type of externalized spatial memory, avoiding, you know, expending energy unnecessarily to revisit patches of food, again, food is being bacteria, food being fungus, and other types of microorganisms. If you can use that to avoid previously forage patches, then this is, this becomes an interesting type of relationship with a particular type of polymucosaccharide, right. And so to test this, this, they have a couple conditions, two conditions. So they have an agar plate, and this agar plate is, the conditions are exactly the same, except for one agar, one plate of agar is covered with extracellular slime, the other isn't. The covered plate is called the coded condition and the blank, the other is the blank condition. They set a U-shaped acetate trap on this particular plate, and the whole idea is to put *Fascerum*, as you see in a start zone, and put a goal which is going to be a drop of highly concentrated glucose, and that goal is food. So if *Fascerum* is going to actually use extracellular slime to navigate, and to avoid, you know, navigating in areas on the agar that it's already navigated on, the hypothesis is that in the coded condition, it's going to take a lot longer, and that's exactly what they found. It took about 10 times the time, actually, it was 10 times greater in the coded condition than the non-coded condition. So to come back to this idea, and here you have kind of the Hans Gretel element, as Reed et al. kind of make the following conclusion. I said, the results of this study provide a unique demonstration of spatial memory systems and non-neuronal organisms, supporting the theory that an externalized spatial memory may be the

functional precursor of an internal memory of higher organisms. I think this is quite interesting, but two hypotheses, obviously, and I think one can safely ignore one of the hypotheses and concentrate on the kind of formal hypothesis about an externalized spatial memory system. Whether or not that system was a precursor to internalized memory, that's something else. Now, in this particular type of conclusion based upon this experiment, memory processes have been, and this externalized memory process has been interpreted as a particular type of cognitive process, albeit one that extends into the environment. Now, is that strange? May? Yeah, sure, why not? It is strange, but really, I think there's a lot of, at least for the last 20 to 30 years, there's been a lot of, you know, a lot of research done and focused upon the idea of what's called, you know, the hypothesis of extended mind, the hypothesis of extended cognition. And, you know, my PhD supervisor, Andy Clark, he was very instrumental, him and Dave Chalmers

were very instrumental in kind of putting that out there. I think that sometimes, you know, cognition can be scaffolded by external elements. And I think this is that, that's fine, I think. If we take that, if we take that assumption on board, it shouldn't really matter whether or not it's on the outside or on the inside, but we'll come back to this. So now I want to think about, in order to actually really move on to the question, is this really memory though, which is a more fundamental question, to think about what memory is. So I'm taking a definition or characterization, as I like to call it, a rough characterization from Francis Pelusky and Michael Levin. And Mike and Francis look at memory as follows in one of the papers. They say it's experience-dependent modification of internal structure in a stimulus-specific manner that

alters the way the system is going to respond to a stimulus in the future as a function of its past.

Now, I think a couple of things stand out in this particular type of characterization. One is that it's stimulus-specific. So breaking a bone, for example, is not going to count as a memory. There are many ways

to break a bone and it doesn't then, the breaking of bone is not going to necessarily be dependent upon one type of stimulus. So that's a very important element of this definition. Another important element, which I'm pushing back on, is the idea that it has to be a modification of internal structure.

Again, if we take the hypothesis of extended cognition seriously,

there's no reason to be dependent upon this assumption about internal structure. Sometimes it extends. And that's an assumption that I'm willing to take on. Maybe some people won't, but that's something that I'm definitely willing to take on, depending upon the functional analysis of what's going on here. Now, putting this into place, you have a different type of characterization. So I think you can add external structure here, internal structure or external structure.

Now, one important thing also about Belusica and Levin's characterization, when they bring this up, they have a footnote which says, you know, memory is going to be outstripped in terms of that memory that's necessarily associated with cognition, necessarily associated with intelligence. So there's going to be many more types of memory than those that are necessarily associated with intelligence or cognition. What does that mean? That's what we're going to get into next, I suppose.

So you have different forms of biological memory that the ones that I'm going to look over right now for this presentation is, these are going to be dependent upon different levels of biological organization, different time scales and biological levels of organization. Now, one is going to be cellular memory, which are individual cells, somatic cells, for example. Immunological memory, networks of cells and molecules, so you're getting a larger, larger time scale.

Individual organisms and then collective memory on a collective scale or group of organisms.

Necessarily, I'm going to look at the notion of kind of colonial memory or colonial memory.

So what's cellular memory? In a nutshell, which I'm sure many of you know, cellular memory is going to describe how various pluripotent cells, you know, stem cells differentiate into different types of cells.

And stem cells themselves, after they start actually differentiating, those cells reproduce.

And as they reproduce, they don't go back to stem cells, and then turn back into, for example, a fat cell.

And the fat cells produce fat cells. So there's some type of way that cells maintain the cellular type over mitosis or cellular division. And this is what cellular memory can be understood as.

One of the ways that this occurs, one of the mechanisms underlying this, again, I hope this is not to simplify it for everyone, but I think it's kind of an important thing to talk about anyway, is through epigenetic modification. And this is one of the ways that epigenetic markers and epigenetic mechanisms were introduced biologically, was looking through or trying to understand the question as to how differentiated cells remain differentiated over time during development, during ontogenesis.

And these epigenetic mechanisms, there are many of them, but a couple of the ones that I have here, just kind of examples of histone modification, where you have, you know, you have an octamer of histones that are going to be wrapped by a particular type of kind of strand of DNA.

Histones are going to make up, these histones are going to make up the chromatin, and going to make up, they're going to make up the larger groupings of cells, sorry, sorry, the larger groupings of kind of DNA.

And when histone is actually wrapped very tightly, when the DNA is wrapped very tightly around histone, it can be accessed for transcription. So what that means, you have some, the breaks are put on to

transcription, so the breaks are put on in terms of genetic processing, which is okay. So that acts in and of itself as a type of memory. And if one is very interested in this, one can look into the work

of kind of, yeah, you say, Yablanka and A.B. Yablanka and Marian Lam, they've done incredible research on this philosophically and biologically. But this is one of the things that kind of, in a nutshell, explains

the way that a cell actually, when it differentiates, remains the same type of cell as the same type of

cell. Now you have histone modification, chromatin marking, RNA interference, there are many different types of epigenetic mechanisms that can be used, but this is just an overview, a very quick and

inaccurate overview probably. Moving on to the immune system memory. Now, I think I noticed that Castro

says it best, so I'm going to read this out for you. Our immune system is composed of a huge number of cells presenting receptors on their surfaces, transmembrane receptors. These receptors are

responsible for binding with portions of pathogens, known as antigens, and signaling other immune cells to eliminate the marked or recognized pathogens. Now, the invading pathogens replicate themselves inside our

bodies, thus increasing the amount of damage being caused to our organism. One way the immune system

evolved to fight against infection was by replicating our immune cells as to cope with the replicating pathogen. But the replication of immune cells is not perfect. You have errors, you have mutations. Now, errors occur with a rate proportional to the quality of recognition between immune receptors and the pathogen recognized. Those mutated cells with high affinity with the pathogen. High affinity means that they have in terms of their kind of their shape, they fit together well. They're selected, maintained, an eruptor called memory. So to just illustrate this very quickly, you have certain cells with transmembrane receptors which are kind of represented by these different shapes coming off these circles, these cells, and they're going to fit better. They're going to have kind of better recognition with some of the foreign antigens. The ones with that are better recognized, and again, they're going to be the ones that are put into a particular type of reservoir, and they mutate, and the mutation ends up actually making transmembrane cell receptors, or transmembrane receptors that are going to actually fit even better with the pathogens. And they're put in the kind of, again, this repertoire of memory, cell memory, or immunological memory. Organismal memory is the thing that everyone typically thinks about, particularly from the perspective of psychology, comparative psychology, cognitive science, or even modeling. And this is just answering the question of how an organism adapts to, via changing experiences, adapts to a changing world over the lifetime. And again, it's over the lifetime that has to be put forward. The idea that something like organismal memory occurs typically, it's going to occur typically over ontogenesis, over a long trajectory of development from birth to death. So the particular type of paradigmatic way of looking at memory and looking at learning also, I suppose, people start looking at this, scientists have started looking at it in Metazoa, just because cognitive science, which is a very still young science, piggybacking off of psychology has started with humans, and then thinking about other types of closely related animals. But now things are obviously changing a little bit, people are asking questions, as Chris Reed has asked questions about memory and physarium. But this is just a little bit of the background. And I'm not going to say too much about organismal memory, because I think this comes quite intuitively given the particular type of type of, you know, our approach to memory and cognitive science already.

Memory, one thing that should be said also is that typically, it's going to be associated with learning. So learning is the way that memories are encoded. Typically, but this is something that we can come back to later, which I think is interesting, particularly with physarium, but with other organisms such as aplica. Now, collective memory, the last type of memory I'm going to talk about, this refers to how history dependent self-organizing behavior of a group of organisms become stabilized at the level of the collective rather than at the level of the individual that makes up that collective. And a lot of the research that has been done on collective memory is the work on colonel memory, ant colonies, for example, or turnite colonies, or these types of insects that have, you know, use pheromones as a manner of navigating a particular type of spatial environment.

Now, Deborah Gordon, she is a fantastic zoologist. And she was at Oxford for a while, and a lot of Deborah Gordon's work was with ants, looking at ants. And she's done some pretty incredible research, kind of looking at ant collectives. And this is something that she writes.

This is a redwood ant. A redwood ant, a colony of redwood ants, remembers its trail system, leading to the same trees year after year, although no single ant does.

In the forests of Europe, they forest high in trees and feed on excretions of aphids, another type of insect, small insect, that in turn feeds on the tree. Now, their nests are enormous mounds of pine needles situated in the same place for decades, occupied by many generations of colonies.

Now, these colonies, I can just interject here, some of these colonies last a very long time.

They have very long life spends, the colony it does, and the colony outlasts the individual ant.

So each ant tends to take the same trail every day, day after day, to the same tree.

During the long winters, the ants huddle together under the snow, and the Finnish

mermicologist Rainer Rosegurtton showed that when ants emerge in the spring, an older ant goes out with a younger ant along that older ant's habitual trail. The older ant dies, and then the younger ant adopts that trail as its own, leading the group, leading the colony to remember or reproduce the previous years' trails. So you have this particular type of connection between nodes, if you will, connecting the memories from one generation to the next, year after year, of the entire colony.

So it's a kind of diachronic connection, if you will.

So after going through all of those notions of memory, I hope that's clear that the different types of memory that are occurring at different temporal and spatial scales, and those temporal spatial scales can be kind of demarcated by different levels of biological organization.

Now we can ask the question, which one of these, or which of these, should I say,

are going to be subject to cognitive explanation as a cognitive process? Now when I say subject to

cognitive explanation, I want to make this clear. I'm asking the question, which of these particular

types of memory systems, or exhibitions of memory, that we should be able to use cognitive theories and cognitive frameworks to try to explore and understand. Now this is not just, for example, using a cognitive theory with scare approach, it's not an as-if explanation, it's not using the intentional stance,

but it's actually using cognitive theories to think about these and identify the mechanisms and functional, the functional placement of these particular types of memory systems as cognitive processes.

So right away, I think it's quite obvious that we do this with organisms. That's going to be the paradigmatic level of biological organization at which we ask these questions, which we apply our which apply our cognitive theories to, our cognitive frameworks to, in a non-instrumental way.

But what about cellular memory? What about immune system memory? What about collective memory?

Again, we do this. How are we going to get, and the question that I'll ask now is how are we going to get any type of, how are we going to explore this question? Because I think it's an interesting

question. We can make some assumptions and just say that there, perhaps there's a, you know, similar principles running through all of them. Given this particular definition of, or, you know,

characterization of memory from Frantysa Beluska and Mike Levin, they all come to memory, so why not apply

all cognitive theories to them? I think there's some, that can lead us into some difficult territory,

and if we do this, a non-sublock to play. And so I want to look at now and introduce a particular type of way of

thinking about this that's going, we can kind of ring a particular type of criterion, a rough criterion, out for thinking about which, which of these biological organizational levels really are warranted for this. Now, warranted to explain in terms of cognitive explanations or warranted targets for our cognitive explanations. One clue here is, it's going to come from this particular kind of passage from Pamela Lyon, and she writes, using, actually, I should say something about this, this particular kind of definition that she offers, or again, a characterization that can be changed. She's offering something that is very much influenced by Sarah Shuttleworth, who's a cognitive ecologist or a cognitive ethologist, excuse me. And this particular definition is, or characterization, is quite, I think, it's again, it's general, but it's general enough to do some work, specific enough to do some work. So cognition here comprises the sensory and other information processing mechanisms an organism has for becoming familiar with, valuing, and interacting productively with features of its environment, exploring, exploiting, evading, in order to meet its existential needs. Now, these are going to be the most basic, which are survival, persistence, growth, thriving, and reproduction. Now, if we concentrate on that, the existential needs, and saying this is fundamental to cognition, we'll have various, you know, we'll have various philosophers, various biologists, various cognitive scientists who will back this up. Maturana and Varela are kind of the obvious ones, people coming from the kind of autopoietic frameworks, autopoietic theory, people coming from the organizational biology theories, for example, people coming from like Hans Jonas, who thinks that, you know, metabolism is the basis of meaningful behavior, any type of self-concern. So any entity that has its own metabolic machinery is therefore this is going to be central to the entities having capacities that are going to be subject to cognitive explanation.

Another person, a philosopher that's out there also, who really emphasizes kind of metabolism and how it kind of shades into cognition is someone like Peter Godfrey Smith.

So, one way to think about metabolism is through the lens of what's called a metabolic individual.

And this is under the kind of, under the larger or more general type of kind of biological, philosophical, a philosophical framework, I suppose, biology, thinking about what biological individuals are.

And this is a question that's come up for many, many, many years. And one of the kind of, I think, fundamental

books about this was written by Julian Huxley, Thomas Huxley, his grandson, and produced and released in 1912.

And it's quite interesting, you know, Charles Darwin wrote a bit also about the biological individual, but it wasn't until I think Julian Huxley's book that a lot of, and he wrote this when he was quite young, I think he was 27.

Many interesting questions came out. He had a very interesting and a keen, a sharp perspective on what the notion of biological individuality is.

Now, what is it? It means we're asking the question about what a biological individual is. We want to provide

an account of what in the biological world constitutes accountable and relatively well delineated cohesive unit. So there's two ways to go about this, two general ways to go about this. There's more than two ways, but I'll just concentrate on two ways here.

You can think about Darwinian individuals, which are going to be units of reproduction. These are going to be any type of units that are capable of

undergoing Darwinian natural selection, which means they have to have, you know, meet three criteria, three criteria, those three criteria that, you know, Avrami was, you know, expressed as, you know, variation, or a heritable variation and selection. So it has to meet these criteria. So you have things like genes, that could be a Darwinian individual, or chromosomes, that could be a Darwinian individual, because these are,

they are, or memes, ideas. These are things that can undergo Darwinian natural selection. These are units.

However, there's another type of biological individual that you typically find in when theorizing about, you know,

various biological processes, various biological phenomena. And this is going to be the metabolic individual.

It's going to be a unit of function that has it, and this unit's going to have it, have its own metabolic machinery.

So let's look now just a little bit more metabolic individuality. I think you can understand how this comes apart from Darwinian individuals, and I'll try to make this more explicit. So, according to Peter Godfrey Smith, metabolic individuals are systems comprised of diverse parts which work together to maintain the system structure. Despite the turnover of material, by making

use of sources of energy and other resources from their environment. Now, diverse parts, that should stand out. They're going to have some type of cohesion that maintains the structure of a system. They're going to be

essentially persists for things that actually serve their own self-maintenance over time. Things that use energy to resist the forces of decay or force entropy only contingently things that reproduce. So that a metabolic individual does not have to be a Darwinian individual. That's quite important. So you'll have certain types of Darwinian individuals. Again, a gene. A gene is not a metabolic individual. A gene does not have its

own kind of metabolic machinery. A gene is not a persister. It doesn't engage in the types of behaviors or activities that keeps itself going.

And metabolic individuality is a quite important concept when it comes to thinking about transitions in the individuality.

But this notion is something for many of you probably know about this, but it's a notion

or it comes from theoretical biology. It's based upon, loosely based upon, veers from the major transitions in evolution.

That was kind of formalized by John Maynard Smith and Errol Sassmeray.

And these, the idea about major transitions in individuality, we're thinking about how one level of biological organization reaches the next level of biological organization over evolutionary time scales. And an example of this might be, for example, an example for this would be going from a prokaryote

to a eukaryotic cell. The evolution from a prokaryote to a eukaryotic cell or the evolution from a eukaryotic cell, a single-celled

eukaryote to a multi-celled eukaryote.

And those particular types of transitions are quite important.

And a lot of kind of biological theories are, you know, focused upon understanding how those transitions occurred over time. What were the circumstances under which they occurred? Did they occur once? Did they occur multiple times?

In many different ways. So, metagraphic individuality is interesting when thinking about, you know, eukaryogenesis, which is how do you go from a, you know, a single-celled prokaryotic organism to a eukaryotic organism with, you know, with a nucleus, with various kind of, you know, cellular parts like a chloroplast or mitochondria.

And the idea here is that there's some type of metabolic integration. Most of the theories that are thinking about kind of eukaryogenesis,

think about this in terms of metabolic integration or two cells, whether it's going to be a bacteria, an alpha bacteria, and some type of archaea, are going to come together in some way. It's going to either be by,

you know, there are different, various theories. One doesn't have to be committed to the theory, but looking at this particular type of illustration,

you have some type of phagocytosis that's occurring from a larger cell to the smaller cell.

When that happens, there's a, depending upon the natures of those two different types of cells, the inner alpha bacteria or the, and the kind of the outer, what is it, going to be an archaea, depending upon that kind of relationship, they could enter into some type of symbiotic relationship such that the output of that endosymbiont, if you will, is something that's metabolically useful for its external environment, which is another cell, and vice versa. And so this nestedness of one metabolic individual in another, after time, is going to, you know, evolve into a single individual, where those two individuals that, and those, the two entities that were at one point in time separate, are now so closely knit metabolically, that they're one type of individual. So you have this idea that metabolic individuals can be nested. And that's, that's a quite, quite important idea here. Metabolic

individuality is going to come in degrees also for this reason. So how might the notion of metabolic individuality hint at what cognitive explanations of memory might get the most purchase? I want to argue the following. If metabolic individuals are first and foremost persisters, and such self-maintenance is to be the fundamental background against which all cognitive capacities might be thought to arise, again, this is an assumption, then the kind of memory, learning, decision-making, anticipation for that matter that warrants possible cognitive explanation, qua cognitive process, is going to be the one that arises at the level of the metabolic individual. Now this automatically rules out genes. When we talk about gene learning, or say gene, gene regulatory systems that are going to actually, you know, be engaged in something like, you know, associative learning or conditioning, which there's some research that shows this, I think we can say, sure, genes do learn. But if that's the case, that gene regulatory networks are not metabolic individuals themselves, I think this is a kind of a misclassification, if you will.

And because gene regulatory networks work within a metabolic individual, they're not the metabolic individual itself. Viruses, for example, viruses are dependent upon their hosts for their metabolism. So these things are not going to, these types of entities are not going to be metabolic individuals. And I think we kind of have a basis for thinking about this, given the assumption that, you know, metabolic individuals are persisters and that cognition, first and foremost, is going to function directly for the entity in which those particular kind of capacities are working for. It's going to function directly for the self-maintenance of that individual. So since metabolic individuality comes in degrees, an interesting kind of question arises. So how much metabolic individuality is going to be required to warrant memory being subject to cognitive explanation and how much is too little? Again, if you remember the idea

that I just said recently, the idea I just brought that you can have cells within cells.

You have a mitochondrion, which actually has its own remnants of its own DNA right now within any type of

somatic cell and somatic cells within organs and organs within, you know, organisms. How do we actually think about degrees of metabolic individuality in this sense that's non-arbitrary?

So I'm going to argue that even if some entity qualifies as a marginal metabolic individual, the fact that it's part of a larger system which even has a higher rank of metabolic individuality suggests that the behavior of the organizational level that's the highest degree of relative metabolic individuality is going to be subject to the explanation of cognitive memory or learning or decision-making.

So the highest degree of metabolic individuality in a nested group, a nested system is going to require actual applications of kind of cognitive theories and cognitive frameworks in a way that's not instrumental or just taking the, you know, the intentional stance.

So we can return to each of these forms of memory and think about applying this particular type of question, applying this particular type of criteria. So we already talked about organismal memory and I think if this is the case, then we can, we can take that off. Sure. Organisms are going to have the highest relative degree of metabolic individuality of their organizational levels of all of the levels that they actually are composed of.

And this is going to, you know, be good news to the notion of asking whether or not say, physarum's externalized memory counts as actual memory. That we should, we should be applying, we can be asking,

you know, whether or not that type of memory, if it's navigational memory, whether it satisfies various types of conditions that, you know, people thinking about navigational memory would need to apply to it to understand its function, to understand the mechanisms behind it better. What about cellular memory?

I want to argue that somatic cells, they have their own metabolic machinery. I think that's quite obvious.

They're, and they're thus, they're metabolic individuals, you know, they metabolize and they have means to metabolize, but relative to the metabolic individuality of the encompassing biological levels at which you find somatic cells, they're quite low. Their survival is dependent obviously upon their environment, but the metabolic individuality, and this is where you get kind of the kind of marginal

metabolic individuality of kind of somatic cells, their maintenance is something that is directed towards the maintenance of the system of which they're a part. So when you get, when you have, you know, somatic cells or groups of somatic cells whose maintenance is reshifted towards their own existence, you start having certain things like tumors. And this is like, you know, metastasis, things of that nature. And I think this is, this is a kind of turning against, this is what you have a kind of turning against, the system of which they're a part. They start using that system and metabolizing that system. They become decoupled from that system. Another way to think about this is in terms of autonomy. When a cell has too much autonomy, when it regains its autonomy from the system of which it's a part, a somatic cell, that cell actually becomes deleterious for the system. So it's not controlled by the system anymore. So I would like to say that something like, you know, cellular memory, we can use the intentional stance to talk about it, maybe that'll get us somewhere. But I think to use something non-instrumentally, is it's actually a conflation. Given, given what, again, given what the different types of transitions and individuality that have occurred over evolutionary time scales have led to the dependencies upon these different types of organizational levels and differences in autonomy. So no, I don't believe that's the case, that cellular memory should be targeted with cognitive theories in any non-instrumental way. Immune system memory, I think it's the same thing, really. The cells of immune system, they have their own metabolic machinery, and a certain degree of metabolic autonomy. Sure. Long-term functioning of metabolic processes and persistence of the network of immune specific cells and molecules that make up an immune system are going to be dependent upon their functional integration with the functional integration with an functional organism. That dependence upon that functional organization is something that is to be expected given where that, where an immune system, where the cells that make up the immune system, how are they related to the larger, the larger encompassing kind of system. So relative to the organism in which you find immune cells or immune systems, those immune cells are not going to actually be more autonomous than that system within which in their worth. Right? So that's that in that sense, if you can start comparing levels of autonomy in that sense, you'll have differences. Of course, these are happening, the autonomy, the actions, the behaviors, if you will, of these systems are going to be happening on different timescales. However, when you're just kind of thinking about whether or not the immune system could ever act in its own accord or for its own accord, that's something that I think gives you a kind of quite direct answer as to whether or not immune systems are going to be, you know, targets for cognitive explanation, which I don't think they are.

Lastly, a collective memory. So the memory at the level of something like a colony. So here's three reasons why I think an ant colony might be viewed as a metabolic individual. It's made up of heterogeneous parts. Again, you have differentiation of colony members. The queen, of course, the reproductive function of the queen, but then you have workers, which are all sisters. I just like bees in the sense. And they have their all the functional allocations, you want to have different types of phenotypes within the colony. And they maintain a particular colonial structure that is going to be dependent over the turnover of individual ants. Now there's a high degree of metabolic integration between these parts. Different ants are going to actually

retrieve food and forage, feed the colony, feed the nests, feed the queen.

And that metabolic integration is dependent upon each of the parts fitting together and functioning together cooperatively. Now there's going to be a small degree of non-cooperation or within ant colonies within any colony, but that's going to be overshadowed by a cooperative behavior. Lastly, you have integration, which is going to be underwritten by external signals like again, pheromones and pheromone trails and behavior. So entering and exiting the nest at a certain type of rate. So it really doesn't, in this case again, like for Xerum, it really doesn't matter if the signals are internal or external. These signals are going to be used. The way that they're used, again, the way they function is going to be analogous to internal signals. So is collective memory subject to cognitive explanation in a non-instrumental way as a cognitive process? I think assuming that there isn't any type of system that is encompassing an ant colony that scores higher than the colony in terms of metabolic individuality. I think an ant colony is going to be in this case where it's going to be maybe equal to the individual ants on occasion. This is a dynamic, obviously a dynamic process, but it might be equal to the individual ants or the individual organism. So colonies come close to having a similar score enough to regard them as something that we can ask the question or apply literally cognitive theories to ant colonies. And I think this is that this kind of buttresses this idea of our colonial intelligence, a colonial memory, colonial learning. So that's that. So in summary, we can return to the question of Xerum again. That was the kind of Xerum is our springboard here. Is it warranted to interpret Xerum's use of extracellular slime, this mucopolysaccharide, as a form of biological memory? I want to say yes. We're warranted in thinking about it in terms of memory. Can such a memory be investigated from the perspective of cognitive memory? Can we apply cognitive explanations to it? I want to say yeah, sure. Xerum is an organism. Organisms are again paradigmatic forms of memory, memory kind of exhibit memory in a very paradigmatic way that we've seen within the talk. And I believe that Fuzerum obviously is going to be a metabolic individual for that reason. So we can ask the third question, is externalized spatial memory a form of cognition? And I say perhaps. I'll leave it at that, perhaps because this is going to depend again, if we can apply cognitive theories, specifically navigational theories of memory and learning to Fuzerum's case, because it is the highest degree, has the highest degree of metabolic individual individuality of its biological organizational levels. That's going to depend upon the different types of navigational theories that we apply. So that third question is going to really be, it's going to be theory dependent. It's going to be framework dependent, whether or not that's going to happen. Michaelian for example, he suggested that these are three different types of criteria for spatial memory and navigational memory. You have to have survival relevant storage, which means that information that gets stored is going to be determined by the relevance of that information to the survival of the individual. Now that again goes very well with the idea of a metabolic individual, sister. Number two, for something to qualify spatial memory, there has to be instability over time. So the obvious way to think about this is memories can be forgotten. They can be overwritten. A memory that can't be overwritten, no matter what type of substrate that memory is in,

doesn't seem to be something like a memory.

It can't be, it doesn't have the dynamics of a memory.

And lastly, there's something like metamemory. He suggests that, you know, spatial memory or navigational memory should be core. This is a core feature of it, which just means that there are other processes that are going to influence whether memory states are acted upon or ignored

in favor of guiding behavior with the current context. And whether or not these all are satisfied

by a physerem, this is a question. But now we get to the point where we can ask these specific types of questions from a cognitive theory, apply it to a particular type of individual, because that individual seems to, you know, kind of, it seems to be a kind of the highest degree of metabolic individuality, or relative degree of metabolic individuality. So in conclusion, regardless of whether

the physerem's use of extracellular slime is considered a cognitive process or not, now again, that's going to be theory dependent, it should be clear that focusing on the behavior of the single-celled non-neuronal organism is going to be quite helpful in revealing interesting and important questions about memory.

And it can help us consider relations between different forms of biological memory.

So again, if cognition is first and foremost a biological process and biological organisms,

approaching this particular, this idea of cognitive explanations using non-neuronal model organisms

like physerem, it's going to offer a really promising way, I think, of going forward, a potential escape from the trap or the, you know, the prison of our own kind of familiarity. And I think that's a really

important thing is to have model organisms that are so different from us that it stretches the imagination, if you will, but stretches the imagination in ways that, you know, expands our horizons and the types of, the different types of questions we can ask, and that help us to kind of reflect on our own case or

cases that are more similar to us. Thank you. Thank you. Awesome. All right. Well, while some people may be

writing any questions in the live chat, I believe the previous time we discussed was in live stream 30.

It was how to count biological minds, symbiosis, the free energy principle and reciprocal multi-scale integration. So some of those topics definitely came back into play, ranging from the metabolic

symbiosis and the multi-scale nesting. Where do you see active inference and the free energy principle in this line of research? Well, I think, I think it's an interesting way to model this, this most certainly

an interesting way to model it. And I think the particular idea of taking metabolism and taking different levels of metabolic integration seriously is something that I don't know, it's something that

I wasn't really, I was thinking about it previously in How to Count Biological Mind, How to Count Biological Science, yes. But I didn't really understand the importance of this as an evolutionary process,

as the importance to it as being kind of the result of an evolutionary process. And once you have these,

taking this into consideration, you do have a manner of kind of delineating where the target's going to be,

and where the most important target's going to be. So you can apply, for example, active inference to any level.

You can apply it to any level, but some levels are going to be more interesting than others.

And I think because it is a type of, you know, active inference, it's a type of process theory,

and it's a type of kind of modeling framework, you're free to kind of choose which levels you want to, you want to apply it to. However, there are going to be some levels that are going to be more, I'm not sure to say this, they're going to, they're going to actually have more of a payoff when you start applying other types of cognitive theories to them. I think particularly when you start thinking about, not literally nesting, but I would say contextualizing those theories within something like, you know, thinking about major transitions in cognitive evolution, or thinking about major transitions more generally, but cognitive theories, putting like that, those particular types of cognitive theories in the context of, you know, evolution. If one thinks about cognition as being something, a kind of suite, or toolbox of various capacities that have evolved. And one doesn't just, you know, take on the assumption that, you know, all life is cognitive, which some people do. But then that question becomes whether or not, you know, it's not a question about how cognition evolved, necessarily,

but it's a question about how life evolved. Those two questions become the same question.

Yeah, awesome. Like, the cognitive turn in evolutionary biology leads us almost to the point of asking, well, what remains other than cognitive account? Is the Venn diagram of the cognitive account, if we take it down to the materials level, ultimately identical with the biological account, like, or the metabolic, all of the substrate of living systems is part of the building up and breaking down of metabolism. So then, you know, what, what is outside of that scope? Like how many different frames and, and, and maybe no need to reduce to one, but point to the, the, the, um, the same cell in the Petri dish? Um, I'll ask a question in the live chat. Okay.

Glea Maximalist wrote, thank you. How do you think the metabolism of different sized organisms relates to their perception of time or their own behavior?

It's an interesting, really interesting question. Um,

I think metabolism can run in an organism and I'm quite sure, I would, I would be quite sure about this claim. Metabolism can run, uh, without an organism being necessarily aware of anything.

It's at a certain level that we can actually have a particular type of, uh, say understanding, if you will, of our metabolism, but metabolism nonetheless will affect, uh, will affect perception. It will affect, uh, kind of how organisms behave, uh, and it will affect probably, uh, their proprioception.

Uh, and, uh, depending upon, uh, because it's, it's about kind of value trade-offs, if you will.

So, uh, if metabolism, for example, if, if, if, if an organism is, uh, in a state that where metabolism is, the function of metabolism is breaking down, there are going to be certain types of, you know, stress responses that occur. That will affect the manner in which an organism behaves. That's quite, that's quite simple. Uh, at which level, you know, organism actually is aware that that's happening. That's a different question. And I'm not, I'm not sure exactly where that happens, that you have to have a quite complex, uh, neuronal system for that to happen. But even before you get to that complex neuronal system, uh, you know, complex nervous system, like even a brain, you want to have certain types of mechanisms in place that are going to be able to, uh, bring an organism back into its homeostatic range. That bringing it back into that homeostatic range is, uh, as something that I would be very certain

that most organisms, if not all organisms have to do in order to survive.

Cool. Also, there's interesting connections, like with the metabolic individuality, you focused on that primary persistence. And then even molecules or processes that are thought of as almost reaching into the semiotic, like neurotransmitters, sometimes are amino acids or just several modifications away. So the line between what's a metabolite and what is a semi-chemical, they are also overlapping because metabolites are key signaling molecules.

Yeah.

Maybe just to your book, like, what do you hope the impact of such a book is, or how did you come to think we need a slime mold and philosophy book and who would read it and think what?

Yeah. Well, I was quite lucky at one point in time.

Uh, I got in touch with Mike Levin and Mike, uh, was, you know, really, he's an, he's an, Michael is a fantastic person. Uh, brilliant guy and fantastic person. And he's like, okay, asked me, okay, like, uh, you've been writing a little bit about, you know, for Xerim, you've been writing about slime mold. Have you, have you got your hands dirty yet? Have you started doing experiments? And I know, not at all. I was like, okay, well let's, let's change that. So Mike sent me some protocol for culturing for Xerim and took me step by step, uh, with a lot of, a lot of communication with him as to how to do this and how to set up some experimental designs. I had been, you know, uh, about a year and a half prior to that, I was over in Rome with Giovanni Pazzullo, uh, learning how to design a kind of behavioral experiments and how Mike was the one that kind of like, you know, reshifted my focus from, you know, human behavioral experiments to behavioral experiments and a petri dish. And the more I got doing these particular types of experiments, nothing published, not, not anything yet anyway. Um, uh, the more I was thinking about what an incredible way to kind of, uh, ask or answer questions that a lot of, a lot of questions that has been happening or arising and kind of theoretical biology or philosophy of biology. You just don't, you don't have, you know, there are a lot of what ifs, a lot of counterfactuals, a lot of, you know, speculation that happens, uh, but given the ease with which this particular type of organism can be cultivated or cultured, given, uh, given, uh, the speed at which it moves, uh, and also, uh, you know, given the, what it actually can do, it's quite unique as to what it can actually, uh, the behavior that it engages in, uh, there are a lot of things that you can proceed to, a lot of questions that you can proceed to ask and just answer. Uh, and so I haven't had the time, uh, or necessarily the resources to sit down for five years and answer all these questions myself with people in the lab. Um, but, uh, at Bochum, uh, Ruhr Universitt, I was given the chance to work with a couple of people, uh, one is, her name is, uh, Chu Rang, and she's, uh, she's a postdoc there and she's doing, she was working on kind of, um, uh, recording climbing plants and doing analysis of climbing plant, um, uh, kind of climbing plant behavior. Uh, you know, I don't know if you've seen, of course, you've seen kind of recordings of plant tendrils as they kind of grow and wrap around vines. Uh, they wrap their kind of tendrils around certain poles and things like this. And she was doing, uh, kinematic analysis of that. And we decided to actually, uh, use that same type of analysis with Viscerum, uh, looking at its growth,

looking at its movement, uh, and in response to certain things. So, uh, I was very, very, very, very lucky to, uh, make her acquaintance and be able to work with her. Um, but I thought, okay, well, prior to this, there's no way that I can do this myself or all of it myself. It's an intensive, it's a really intensive research program. There's so many questions that can be asked. And I think part of the, part of the fun of it, uh, and I think part of the value of it is asking the questions and maybe inspiring people who want to, want to answer these questions in the lab, uh, to give them a couple kind of questions to kind of, uh, you know, to use as a fulcrum, if you will, uh, and to get philosophers of biology, philosophers, philosophers of science, uh, and biologists, microbiologists together in one space, uh, to work together and perhaps think about these questions and, and use resources, uh, you know, to answer these questions, if you will, uh, in a laboratory setting. And so that's what the book is about pretty much. It's like, it's looking at how *Vesicularia*, which is just like, it's, it's an organism that I'm absolutely, you know, I'm, I continue to be mesmerized by, you know, uh, the, the various things that it can do and just how, how beautiful of a, of a life cycle it has, uh, and interesting of a life cycle that it has, uh, how it can be used to answer some questions from very, very longstanding questions in the philosophy of biology and say philosophy of cognitive science also. Awesome. That resonates a lot. I'll add a few points. I'll begin with a quote from a 1989 paper, a framework for plant behavior by Silvertown and Gordon, who you quoted. And they wrote, there's some question whether words used to describe animal behavior, such as the word behavior itself or foraging can be properly applied to the activities of plants. So it's kind of this evergreen discussion. Oh, okay. But is it really solving that maze or is it really memory? And certainly Mike Levin and, and others, and these come up, like is anything that's outside of the

organismal frame actually doing cognition.

And then the slime mold blurs that because it's also a single cell.

So then it's like, well,

why would it matter if there's a lipid membrane separating this or that when the cell is also the organismal boundary?

And then also to the relevance and sort of the cross fertilization of philosophy of biology and the behavioral sciences or ethology in professor Gordon's undergraduate class.

One thing that she would have the undergraduates do was design and carry out a behavioral experiment. Even, you know, do squirrels cross the bike path more on Wednesdays?

And there's something about doing any inquiry that provides philosophical insights that, that the armchair doesn't.

And it helps reveal there's an associative analogical mode.

That's like studying foraging across different systems.

However, by virtue of thinking and scoping over so many systems, most are not going to be directly engaged with, you know, we're thinking about like a space hypothetical asteroid forager and, um, ants as a broad category.

But then when you get into the natural history and the behavioral science or the constraints of a specific laboratory, just sticking with one system like a drosophila or slime mold or this one field site, those can offer so many questions that even leads to reinterrogation of what the bigger category like navigation was.

So for example, like in the, in the navigation example, or the maze solving, I thought, well, it's doing something spatial, but not necessarily with a cognitive map.

So what is, does now it's like,

so actually there's some play between what is cognition that's realized spatially and that, which is aware of its own spatiality.

And that's something that can just be brought to bear with the Petri dish and some experiments like that. So it's just when the philosophy questions are held up with the inquiry, there's just amazing possibilities.

And like your, what you shared shows that, you know, people come from both sides and, and, um,

may it be a perforated membrane between the two because there's so much to like learn on both sides. Absolutely.

Absolutely.

Absolutely.

I think this is like a, you hit the nail on the head.

It's just, um,

I take it then that you were,

you were in, uh,

Deborah Gordon's class then you did you take courses with her?

She was my PhD advisor.

Oh,

of course.

Fantastic.

That makes sense.

It makes a lot of sense.

She's incredible.

She's really incredible.

Um,

so yeah,

I think that you've,

you hit the nail on the head with many,

many of the things that you just said,

um,

in terms of say spatial navigation,

uh,

yeah,

for Xerum,

not going to have a cognitive map,

not going to have a cognitive map.

And this is,

I think like,
uh,
I think cognitive maps are quite complex things that there's still disagreement,
as to whether or not,
you know,
uh,
which organisms use cognitive maps.
Uh,
again,
Sarah Shuttleworth,
2014,
she really kind of puts the brakes on and said,
if you can explain this,
a lot of what organisms do,
animals do,
she's a,
she's an ethologist,
uh,
in terms of using certain types of,
uh,
you know,
other simpler mechanisms,
uh,
that they're still,
they're still quite complex cognitively,
but they're simpler mechanisms and cognitive maps,
uh,
running computations on cognitive maps.
Um,
uh,
you have,
you have these things going on in ants,
for example,
uh,
where you have,
uh,
like,
uh,

egocentric or,
uh,
versus,
uh,
allocentric kind of recognition of landmarks and things of that nature,
or,
um,
you probably know this,
the term better than me,
but dead reckoning,
uh,
using dead reckoning.

Dead reckoning is distinct from using a cognitive map.

It probably can be used in addition in animals with cognitive maps.

Uh,
but the question that arises in the baserum,
is there something similar to dead reckoning happening?
Is there something similar to landmark navigation happening?
Is it leaving its own landmark?
Something like this,
you can ask these questions,
start to ask these questions,
uh,
because you can relate its behavior,
uh,
to other types of cognitive theories.

If you have the right,
if you're asking the right level,
if you're at the right level,
you know,
uh,
so yeah,
it's not going to satisfy any,
any of this kind of criterion,
if you will,
for thinking about cognitive maps,
but maybe landmark use,
perhaps,

you know,

landmark use does not necessarily have to be a personal process.

It can be a self-personal process.

I think it's a Hofstetter quote,

like computers don't play chess.

They do statistics,

but they don't do statistics.

They just follow the laws of physics and all of that.

It's like cognitive phenomenon.

And then we start to really interrogate them and break them apart.

Oh,

well that outfield,

there's not really inferring the location of the ball.

They're just using a behavioral heuristic to stand where it's going to,

it's like,

well,

if that's actually how it's done,

then what is the space between the problem as it was set up and how it's really being done in all these different biological ways.

So,

and that even came up in the reciprocal integration.

Like,

what do we expect?

What are we surprised by?

What,

what does what we're surprised by say about who and how we think we are in how we analyze these cognitive systems?

Um,

I'll ask a last,

last,

last question from the live chat.

Upcycle club wrote,

how might the adaptive behaviors of slime molds contribute to our understanding of distributed cognition?

Okay.

That's a good,

that's a good question.

I think it relates back to,

I would have,

I would have to say,
uh,
really very important work by Chris Reed,
uh,
and his colleagues over at Macquarie university.
The first paper that I referred to,
to the 2012 paper about externalized spatial memory,
uh,
or externalized navigational memory is,
I think that's a good place to start with that in terms of,
uh,
an organism that uses its own byproducts to navigate.
Uh,
it uses its own byproducts.
Uh,
you know,
again,
this is not,
this is not a personal level kind of choice or anything like this,
but I should say this,
I think this is quite important in terms of understanding the nature of it.
Once we get to say,
okay,
well,
this is the right level for looking at memory.
This is perhaps the right level for looking at the organismal level or,
or biological organizational level for thinking about things like problem solving,
decision-making and so on and so forth.
It's use of extracellular slime,
which again is a mucopolysaccharide.
It can be overridden.
So it's not just a repulsion to this stuff.
Uh,
if you create a situation,
uh,
in a Petri dish where you have a circle of extracellular slime,
this is one of the experiments I've done,
and you put a fuzerum in that circle,

uh,
and then you put a reasonably sized oat flake,
which that's something that you use as kind of,
um,
uh,
uh,
uh,
it's for food in order for motivation.
Um,
it will,
it'll climb over the extracellular slime.
It'll,
and that's one of the things that I think is quite important,
uh,
and thinking about,
and that was one of the kind of criteria also about kind of spatial memory or navigational memory,
uh,
that was brought up in the talk,
uh,
that something can be overwritten given the particular type of context.
I would think there are many other ways that you can expand this.
When you can ask the question,
when does it use this extracellular slime and what does it use it?
It's extracellular slime for.
Is it,
is it just,
uh,
you know,
again,
is it just a lubricant for,
uh,
motility?
No,
we know that it's kind of uses it for navigation.
What,
in what ways does it use it for navigation?
And does it always have to be in contact with it?
Is,

are there gradients?

Are there,

there's,

there are many questions that could be asked,

but then I guess the most important thing to think about is the fact that whatever's going on,

it's,

it's a coupling of a cell to something in its external milieu.

And depending upon the geometry of that extracellular slime,

depending upon the values of various stimuli within its surrounds that it can sense again,

it's,

it's,

it's sensing these various,

it's sensing food,

for example,

uh,

through concentration gradients,

uh,

and it,

through,

through agar that are going through the air,

whether it's going to be oats or sugar from oats that are kind of leaking,

seeping into the agar,

or whether it's in its own natural environment,

uh,

it can do this particular type,

these,

I'll call them computations,

if you will,

on,

uh,

what's available and what's more important,

uh,

for this particular type of context.

Uh,

and extracellular slime plays a,

plays a role in that,

in the sense that,

uh,

it's one of the things that it weighs in these particular types of computations.

The computations,

again,

might be very,

very,

very,

very simple,

uh,

but they're,

they're computations that are being run because it's in the integration of information.

And the weighing out of this information,

according to,

uh,

according to kind of the continuation of metabolic functioning,

when things don't work out,

then it goes into one of these dormant stages.

And it can reemerge after waiting things out.

But that's,

uh,

extracellular slime plays a very,
very important role.

So again,

the suggestion,

look at the work of Chris Reed,

um,

read it out,

2012,

2013.

And it's a very good starting place to understand how *Fuzerum* uses extracellular slime in these ways.

I'm also struck by when you showed the life cycle diagram and how it seems like less is known about what cognitive tasks or work is done in those other stages.

And then also how different they looked from a size and morphology.

And it just is like,

what,

what we,

you know,

the,

the caterpillar and the larva and the butterfly all do different things.

And so when we think about the life cycle perspective and community,
there's so many cool ways to go.

Well,
if you have any last comments,
please go for it.

Otherwise it's a great direction of research.

And it's exciting to see how the cognitive sciences are entering the chat and bringing a lot of excitement to
these perennial questions about individuality and cognition.

I think so much.

Yeah.

I think in terms of life cycles,
I think there's something fun that,
and very important that goes on.

One of the things that has beset,

I said,

I suppose,

biological research,

a lot of biological research and specifically,

you know,

thinking about cognition is that we're adults' interests.

We think about the adult organism,

you know,

of course there's developmental psychology and kind of people are looking at that,

but most,

a lot of the research,

when we're talking about generalizing certain types of capacities and generalizing certain type of,

you know,

response rates,

we have a very kind of narrow band of model organisms and ages of model organisms that we're looking at.

this even,

when you look at an organism with a complex life cycle,

you expand that in such an incredible way because you're not just looking at different ages,

you're looking at different,

really,

really different phenotypes within the same organism over,

over,

over time.

And not only are you looking at different phenotypes,

you're looking at,
and this is one of the things that I approach in the book,
it's a wonderful way to explore something like niche construction.
Because if those different parts of the life cycle are responses to changes in an environment,
if they're responses to various selection pressures,
ways to accommodate selection pressures,
sometimes they might happen,
they might be primed by certain types of stimuli,
sometimes not,
sometimes they might be primed by certain types of things.

then you have a different kind of frame
that one can use
to think about niche construction,
various types of niche construction and how they're related.
So yeah,
a beaver,

you know,
the stereotypical example of the beaver building a dam,
to kind of control the dynamics within its particular type of niche.
You have that,
like that's a particular type of externalized niche construction,
that's a particular type of niche construction.
But then you have other things which involve, for example,
an organism changing its own morphology,
as a way to kind of buffer itself from changes in selection pressures,
harder selection pressures.
So a change in temperature,
it might be a change in, for example, moisture.
So the mix amoeba,
then,
you know,
it will synthesize flagella,
become a bi-flagellate,
if there's,
if it starts raining,
if there's a lot of water,

then those,
it'll stop synthesizing those flagella and start crawling again as an amoeba,
as if it dries out again, right?
And there's,
there are kind of costs,
like the costs of,
of that particular type of morphological change,
or that type of niche construction,
are quite interesting things to think about.
Because when it's a bi-flagellate,
there,
it's,
it can't reproduce.
So there's a trade-off there.
When it's a mixed amoeba,
when it's reproducing,
it can't feed.
So,
there,
there,
it's very,
it's very,
it's very,
it's very,
it's very,
interesting way to think about,
life history trade-offs,
plus niche construction.
And you get all of that,
with an,
if you focus rather than on,
just the,
just the plasmodium cell that we've been talking about during this,
that we've been focusing on,
on this talk,
on the entire life cycle.
And there are many other organisms,
bryophytes,

for example,
ferns,
you know,
incredible,
incredible life cycles that,
you know,
they,
they,
behavior is different in,
and obviously in,
and bryophytes or plants or whatever,
but the,
the same goes for them in the sense that,
you have,
that,
that life cycle provides a really beautiful way,
a beautiful lens through which to see,
how an individual's morphology is very much coupled in,
in,
in incredible ways to different environmental dynamics,
and how it can actually influence those dynamics,
and influence its own evolution.

Epic.

Okay.

Thank you,

for joining again.

Thank you so much.

And looking forward to reading the book.

And until next time.

Thanks,

Bye.

Bye.

Bye.

Bye.

Bye.

Bye.

Bye.

Bye.

Bye.

Bye.

Thank you.